WHEN TRUST MATTERS

Energy Transition Outlook: Hydrogen Roadmap to 2060

The Future of Hydrogen in Shipping

Chang Woo SON, Deputy Head of CMC Service Line

01 September 2026

Agenda

Cost first, then the regulation that closes the cost gap, then demand and supply — and finally what it means for ships.

Why Hydrogen Matters

The last piece of the Net Zero puzzle

Cheap solar, wind and batteries have advanced faster than expected — and have pushed hydrogen out of buildings, medium-heat industry and even heavy trucks. What remains is the set of sectors that cannot be electrified at all.

DNV key message

Hydrogen is not competing with electricity. It completes what electricity cannot do.

HARD-TO-ELECTRIFY SECTORS

Deep-sea shipping

Batteries suit ferries; energy density rules them out for deep sea

Aviation

Weight and volume constrained — drop-in liquid fuels only

Steel

H_2-DRI is the only large-scale near-zero primary steel route

Chemicals & fertiliser

Hydrogen is the feedstock itself, not an energy option

High-temperature heat

Above the range electrification can reach economically

Today's Hydrogen Industry

Hydrogen is not a new industry — it is a large, mature and highly carbon-intensive one.

USD 240 bn
Annual market value

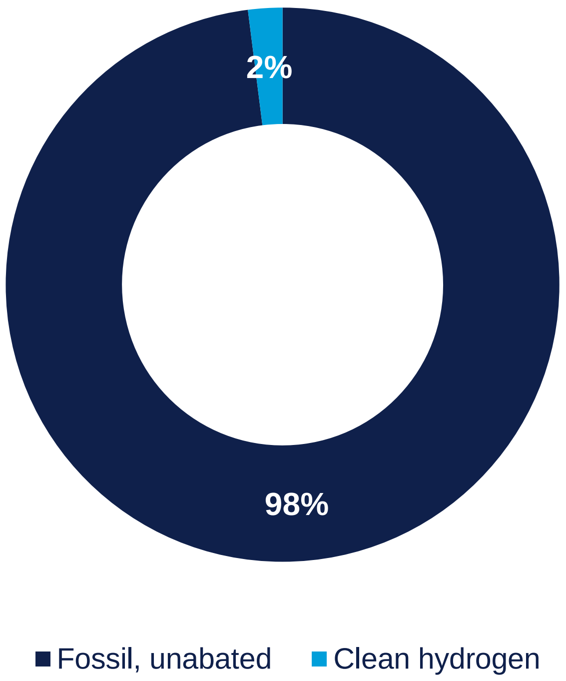
~100 Mt/yr
Hydrogen produced

1300 Mt CO_2
Emitted every year

The hydrogen industry is itself a decarbonisation target.

About 98 % comes from unabated fossil routes — mainly steam methane reforming (SMR) and coal gasification. Clean hydrogen supplies less than 1 % of the global market today, rising to 5 % by 2030 and 10 % by 2035 in DNV's forecast.

Global hydrogen production mix, 2025



From Hydrogen Hype to Reality

DNV 2050 clean hydrogen forecast

−45 %

revised down between the 2022 and 2026 Outlooks

A filter, not a collapse

DNV reads the cancellations as the pipeline filtering down to projects with credible offtake. Hydrogen remains indispensable for shipping, aviation and steel — the timeline moved, the requirement did not.

WHY THE FORECAST WAS CUT

1. **Announced pipeline shrank**
   From 50 Mt to 37 Mt of announced projects between 2024 and 2025
2. **Few projects reach steel in the ground**
   Over 1,500 projects planned; about one third at FID*; only a handful in construction
3. **Policy support stalled or reversed**
   Most visibly in the US, but not only there
4. **Electrification moved faster**
   Cheap PV and batteries took segments once assumed to be hydrogen's

* FID: Final Investment Decision

Hydrogen Terminology

The industry is moving away from colour labels toward definition by measurable carbon intensity.

Grey Hydrogen

- Steam methane reforming (SMR)
- Coal gasification
- No CO_2 capture
- 98 % of supply today

Clean Hydrogen

Blue / Low-carbon

- Natural gas feedstock
- With carbon capture & storage
- Capture rate is decisive
- ~20 % of 2060 supply

Green / Renewable

- Water electrolysis
- Renewable electricity
- Near-zero carbon intensity
- ~70 % of 2060 supply

Clean Hydrogen

DNV definition

= Low-carbon + Renewable

By 2060 clean hydrogen is about 90 % of production — some 240 of 280 Mt/yr.

Hydrogen Derivatives

Derivatives are about half of all clean hydrogen uptake from the mid-2030s to 2060

NH_3

Ammonia

Carbon-free carrier with a dual role as fuel and fertiliser feedstock. Existing global trade and storage.

CH_3OH

Methanol

Liquid at ambient conditions. Ten-year track record as a marine fuel; simplest retrofit.

e-CH_3OH

e-Methanol

Green H_2 plus biogenic or captured CO_2 — and that CO_2 is the binding constraint.

e-SAF

Synthetic jet fuel

Drop-in for existing aircraft. Competes with shipping for the same molecules.

Why derivatives matter

- ✓ **Ambient-condition storage**
- ✓ **Existing seaborne trade**
- ✓ **Bunkerable at scale**

Ammonia takes the largest share of derivative production, carrying both the fuel and the fertiliser demand.

Why Shipping Needs Derivatives

Pure hydrogen at sea

- Low volumetric energy density, compressed or liquefied
- Cryogenic storage and boil-off over long voyages
- DNV: large-scale adoption in international shipping unlikely
- Relevant to shipping mainly as cargo, not as bunker fuel

Ammonia and methanol

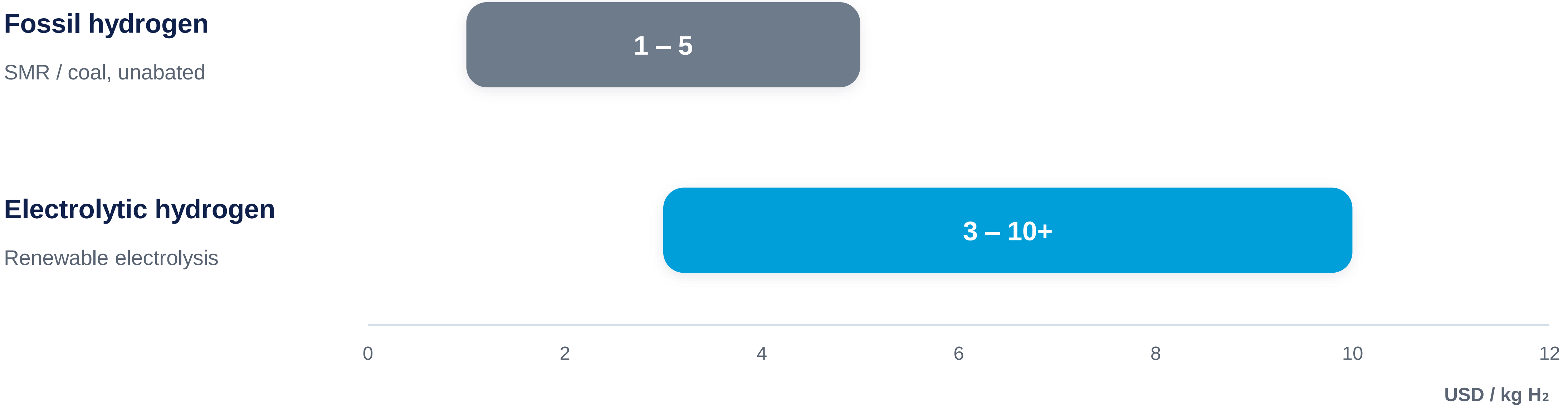
- Hydrogen embedded in a denser, storable molecule
- Established seaborne trade and terminal infrastructure
- Bunkerable at concentrated hubs — traceable at delivery
- Engine and retrofit technology entering the market now

Almost exclusively

On DNV's assessment, virtually all maritime hydrogen demand shows up as a denser molecule carrying the hydrogen — because that is what can realistically be stored, shipped and bunkered at fleet scale.

The Cost Gap Today

Levelised production cost, current ranges (USD per kg H_2)



~USD 5 /kg
Global weighted average for renewable electrolysis today

USD 3.7–4 /kg
Methane reforming with CCS — the blue benchmark

Only China
has consistently pushed the range lower, on cheap technology and capital

How Far Costs Fall

Electrolyser capital cost (non-Chinese, before EPC premium)

USD 880

per kW, 2025

USD 539

per kW, 2060

Via USD 737/kW by 2029 and USD 650/kW by 2032 — each step one doubling of cumulative installed capacity, at a 15 % learning rate declining to 12 %.

USD 350 /kW

Chinese alkaline electrolysers, held flat across the whole period — an already mature manufacturing base. Offset partly by 15 % more frequent stack replacement.

WHAT ACTUALLY DRIVES THE COST DOWN

Power price, not capex, decides

Electricity dominates the cost of renewable hydrogen. Average PPA prices fall from ~USD 65 to ~USD 40 per MWh by 2060.

Renewable LCOE* falls 45–55 %

Solar and onshore wind cost declines feed straight through to hydrogen.

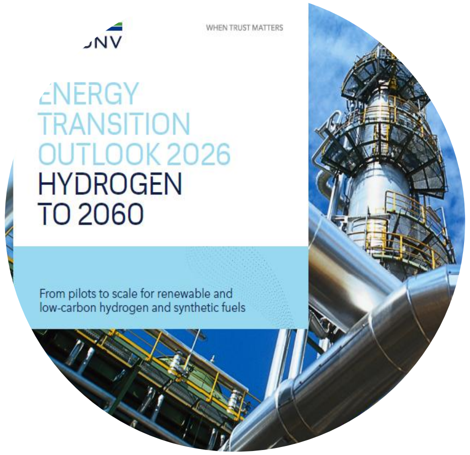
*LCOE: Levelized Cost of Electricity

Higher capacity factors

More annual operating hours per installed kW improve capital recovery.

Result: off-grid renewable hydrogen falls to about USD 2.6/kg by 2060, and below USD 2/kg in the best regions — but not everywhere, and not soon.

The Gap Does Not Close by Itself



Clean hydrogen will not be commercially competitive on its own, and DNV expects it to stay that way until emitting carbon becomes expensive enough.

DNV Energy Transition Outlook 2026 — Hydrogen to 2060

Both sides of the gap have to move — and neither moves on commercial logic alone.

Supply side

Learning rates and cheap power narrow the gap, but do not eliminate it within the forecast period

Demand side

Shipowners compete globally and have little incentive to absorb higher fuel cost without binding rules

Therefore

Regulation is the mechanism that eventually closes the cost gap — DNV's own formulation

Shipping Regulation — IMO

IMO ambition

Net Zero by 2050

GHG emissions from international shipping

THE INSTRUMENTS

1. **IMO Net-Zero Framework**
   A first-of-its-kind carbon pricing mechanism requiring ships to reduce GHG intensity progressively against a historical baseline.

2. **2024 IMO LCA Guidelines**
   Default well-to-wake lifecycle values, already in force — the accounting layer that makes any fuel claim auditable.

Not yet adopted — earliest effect 2028

The NZF has not been adopted, owing to resistance from several member states. DNV's reading: this does not change the long-term direction, but it creates near-term regulatory uncertainty, and many owners are hesitant to commit to new fuel technology until the framework is confirmed. For the classification interface the problem is sequencing — vessels ordered now will trade deep inside the regulated period, so the fuel-ready or fuel-capable decision has to be made before the rule text is final.

European Regulation

Already enacted — these are the rules generating cost today, ahead of any global instrument

EU

EU ETS

In force since Jan 2024

Extended to ships of 5,000 GT and above entering EU ports. Emissions carry a direct allowance cost.

EU

FuelEU Maritime

Well-to-wake intensity

A tightening GHG intensity limit on energy used on board, with pooling and banking. Non-compliance is priced.

EU

RED III

Binding RFNBO[1)] levels

Minimum renewable fuel of non-biological origin shares for industry and transport by 2030 — the demand instrument behind e-fuels.

CBAM[2)] on hydrogen imports took effect in 2026, and the Commission opened an early review of the RFNBO rules in April 2026 — the European framework is still moving.

1) RFNBO: Renewable Fuels of Non-Biological Origin

2) CABAM: Carbon Border Adjustment Mechanism

Certification & Carbon Intensity

For clean hydrogen, proving the carbon intensity is harder than producing the molecule.

1

Carbon Intensity

gCO_2e per kg H_2 on a defined system boundary — well-to-gate, well-to-tank or well-to-wake depending on the scheme.

2

Certification

Independent verification of production route, power source and capture rate. Korea's Clean Hydrogen Certification Scheme is tiered 0–4 on a well-to-gate boundary.

3

Traceability

Chain of custody through transport, conversion and bunkering. Concentrated fuelling nodes make point-of-delivery checks feasible.

Where class comes in

DNV notes that regulation cannot function until the standards and recommended practices beneath it exist — and that the absence of mature standards is one reason the market has formed more slowly than forecast. Third-party verification of carbon intensity, and of the safety case that goes with it, is the classification interface to the hydrogen value chain.

The Instruments That Close the Gap

Four mechanisms do the work — and the money behind them is already committed

1

Carbon Price

Prices the fossil alternative. EU ETS today; the IMO pricing element next.

2

Subsidy

Capital and operating support. EU: over EUR 20bn allocated across programmes.

3

Contracts for Difference

Japan's GX allocates about USD 20bn in CfDs to producers and importers.

4

Mandate

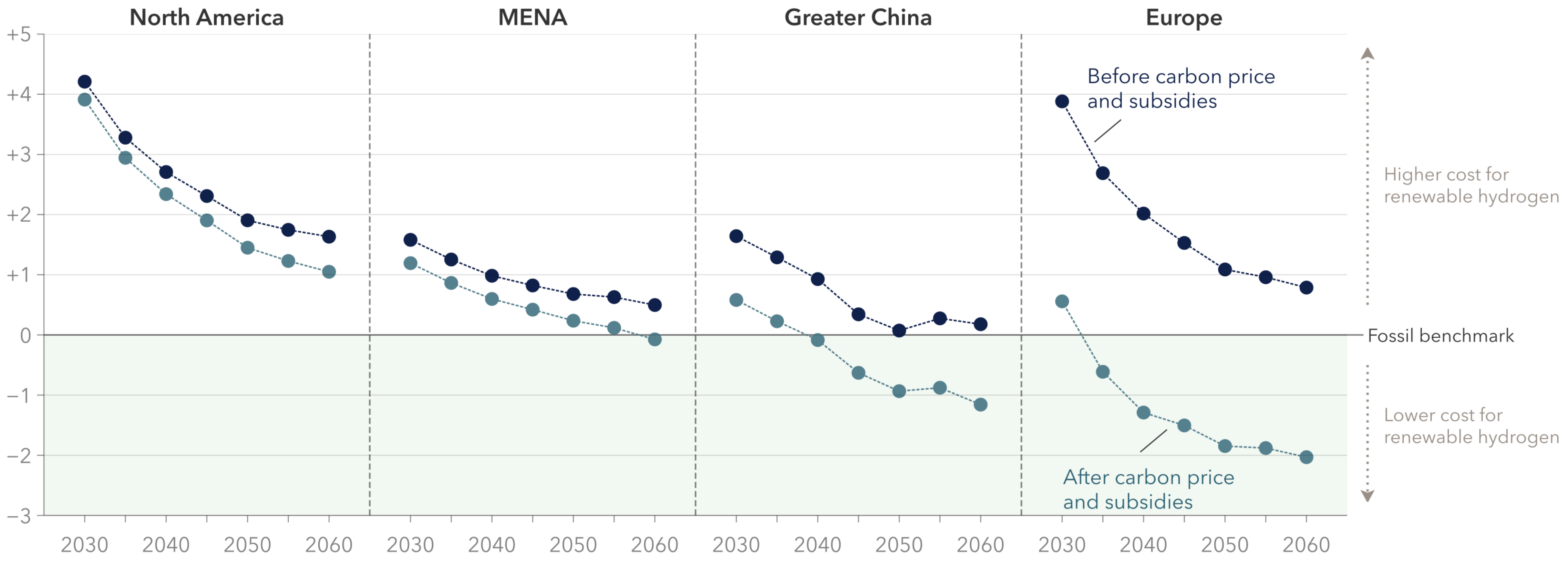
Obliges use regardless of price — RED III, FuelEU Maritime, ReFuelEU Aviation.

Asia-Pacific

Under the CfD framework, South Korea has set a target of 27.9 Mt/yr of clean hydrogen by 2050, of which about 80 % is expected to be imported — alongside the Clean Hydrogen Portfolio Standard. Japan targets 3 Mt/yr hydrogen and 3 Mt/yr ammonia by 2030, rising to 20 and 30 Mt/yr by 2050. These two demand signals are what make the Asia-Pacific trade corridors bankable.

Regulation and policies are crucial for further scaling

Cost gap between unabated fossil and renewable hydrogen (USD/kg)



Where the Demand Actually Sits

DNV groups future demand into three distinct blocks — the shape matters more than the total

THE FLOOR

Legacy industrial base

Fertiliser, methanol feedstock and refining. No alternative exists, so these are the earliest and most certain adopters of clean hydrogen.

THE GROWTH

Transport fuels from hydrogen

Maritime and aviation together are the largest source of new demand in the long term — almost entirely as derivatives.

THE DISAPPOINTMENT

New pure-hydrogen uses

Steel dominates what remains. In road transport and heating, direct electrification simply wins on cost.

Planning implication

A narrower demand base means the clean molecules that do get produced will be contested. Shipping competes directly with aviation for limited sustainable biomass and synthetic fuel — DNV names this explicitly as a constraint on maritime uptake in the 2020s and early 2030s.

Hydrogen Demand by Sector

Largest sources of new demand — hard-to-electrify sectors only

Maritime

Ammonia and e-methanol as bunker fuel

Aviation

e-SAF complementing bio-SAF

Steel

H_2-DRI where scrap recycling is insufficient

Fertiliser

Decarbonising ~190 Mt/yr of existing ammonia

Chemicals

Methanol and other feedstock uses

Common denominator: **no electric alternative at the required energy density or temperature.**

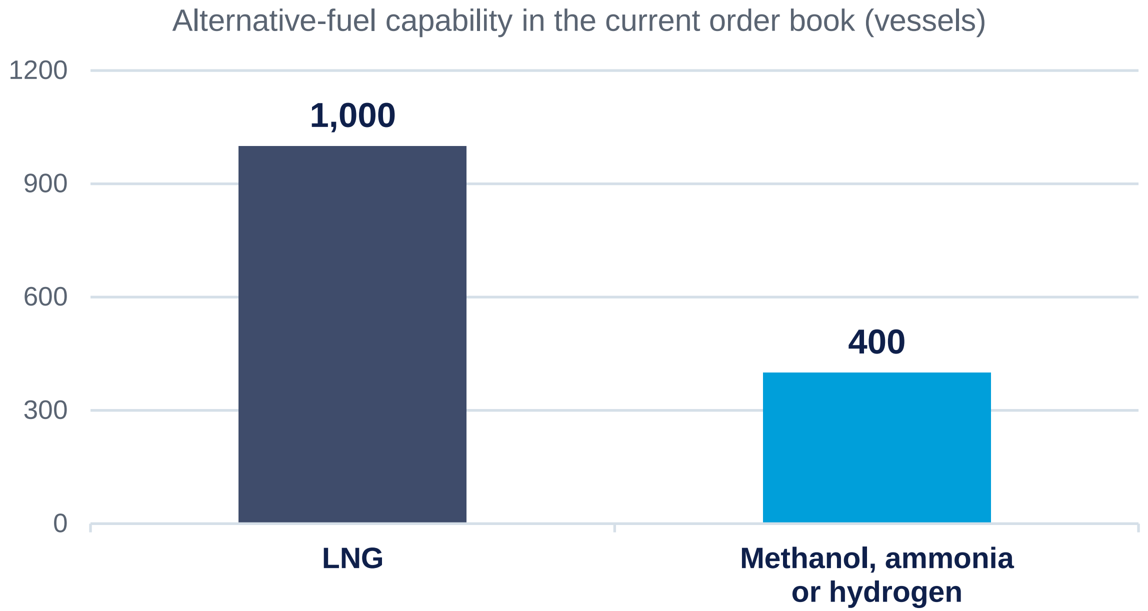
Maritime Demand — the Numbers

How much hydrogen the fleet actually needs, and how ready it is to take it

~15 % of global hydrogen demand across all sectors, where maritime stabilises from the mid-2040s

~40 Mt H_2-equivalent per year implied at that share of a 280 Mt market in 2060

Largest new demand source in the long term — maritime and aviation together



About half of the gross tonnage on order has alternative-fuel capability — but the order book is still dominated by LNG.

Fuel-ready is not fuel-capable: only the latter generates real demand.

Maritime Fuel Transition

Efficiency first, fuels later — and then ammonia overtakes methanol

2020s

Efficiency and biofuels

Slow steaming, routing, hull optimisation and wind assistance deliver the near-term cuts. Drop-in biofuels serve the existing fleet with little or no engine modification.

2030s

e-Methanol leads

Easier handling, simpler retrofit and a ten-year track record as a marine fuel. Growth is capped by the availability of biogenic or captured CO_2.

2040s

Ammonia dominant

Once engine readiness and safety standards are resolved — DNV expects the early-to-mid 2030s — ammonia scales faster than e-methanol and becomes the dominant contributor in the early 2040s.

The brake is structural, not ambition: slow fleet turnover, bunkering infrastructure, fuel certification and port readiness — all of which sit in the classification and survey path.

Steel — the Other Anchor Demand

Relevant to shipbuilding twice over: as a competitor for hydrogen, and as the source of the plate we build ships from.

H_2-DRI

Hydrogen Direct Reduced Iron

DNV's assessment: the only large-scale pathway to near-zero-carbon primary steelmaking where scrap recycling is insufficient. Steel is the largest single source of long-term pure-hydrogen demand.

What holds it back

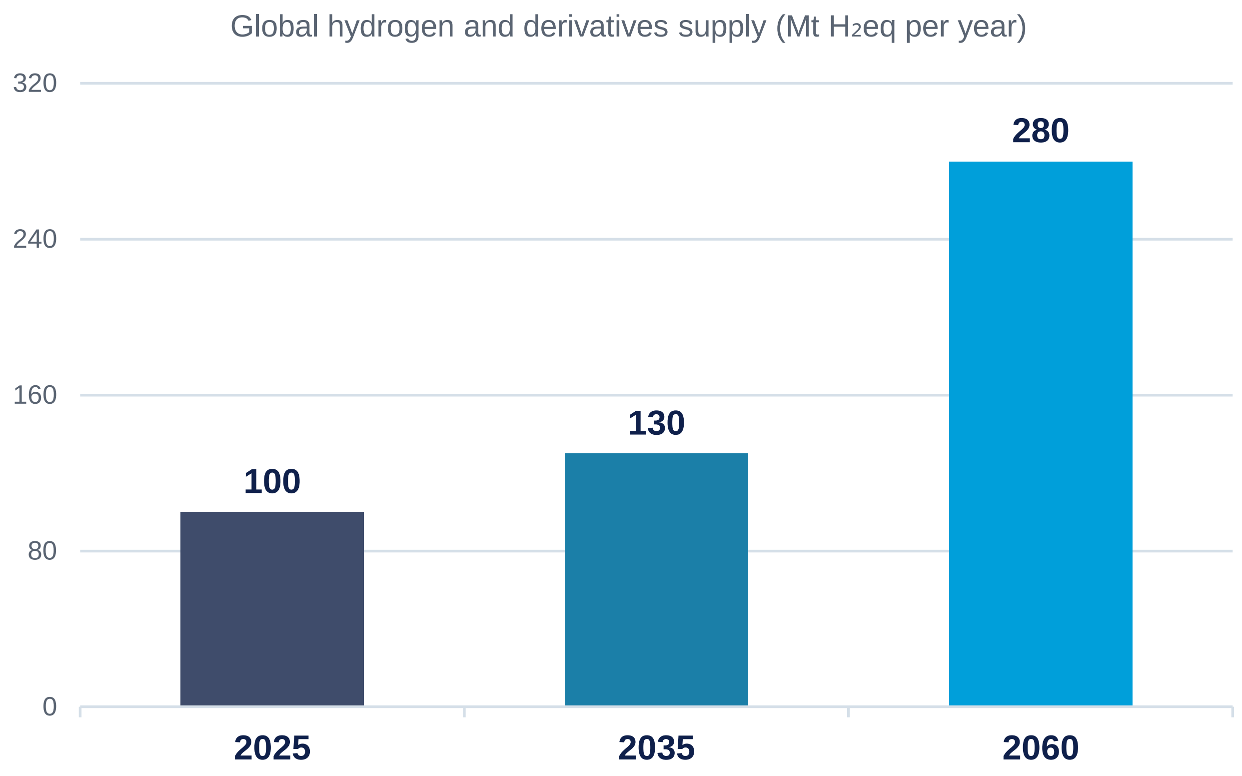
- High hydrogen cost against incumbent coal
- Long blast furnace lifetimes; relining decisions are generational
- A relatively young, efficient BF-BOF fleet in parts of Asia
- Uptake described by DNV as delayed and uneven

Into the 2040s

DNV expects existing blast furnace capacity to limit the role of hydrogen-DRI well into the 2040s.

Hydrogen Production Outlook

Volumes grow steadily; the transformation is in the mix, not the total.



100 ×

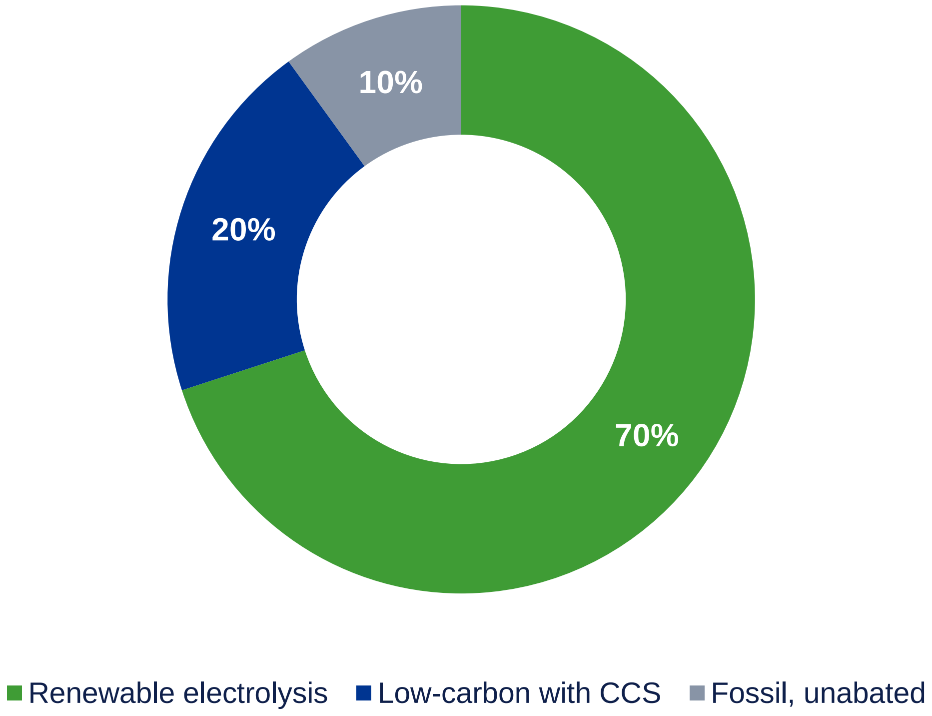
growth in clean hydrogen between today and 2060, reaching about 240 Mt/yr — roughly 90 % of all production.

Demand-led, not supply-led

DNV only counts projects with identifiable offtake: if demand does not materialise, the capacity is not built. Roughly 20 % of announced nominal capacity is assumed to materialise by 2035.

Production Mix in 2060

Hydrogen and derivatives production by route, 2060



70 %

Renewable electrolysis

The dominant route. Grid-connected electrolysers on PPAs produce about 60 % of it; dedicated off-grid renewables the rest.

20 %

Low-carbon with CCS

A complement, not a competitor — concentrated where gas is cheap and CO_2 storage exists.

10 %

Fossil, unabated

Down from 98 % today. About two-thirds of the remainder is natural gas reforming.

Clean hydrogen reaches about 240 Mt/yr — roughly 90 % of all production.

The Rise of China

China is becoming the centre of gravity of the hydrogen industry — as it already is for solar PV and batteries.

62 %

of global electrolyser manufacturing capacity by 2024, up from below 10 % in 2020

USD 350

per kW for Chinese alkaline units — against USD 880 elsewhere today

~35 %

of all new hydrogen production and use over the coming decades

Implication for Korean shipbuilding and shipping

DNV's own reading is that China is repeating the state-directed playbook it used for solar and batteries. Cost leadership in electrolysers shifts where derivative production gets built — and therefore which trade routes and which carrier designs the market will need. Chinese equipment carries its own qualification questions: certification, performance guarantees and bankability are named explicitly as open issues.

Trade Flows and the Asia-Pacific Corridor

Trade organises around regional corridors rather than a global market — and Korea sits on the receiving end of the largest one.

AUSTRALIA — THE PIVOTAL SUPPLIER

~2/3	of all regional hydrogen trade runs Australia → Japan, Korea and Singapore
>80 %	of Asia-Pacific hydrogen and derivatives traded internationally by 2050
MENA	exporting over 8 Mt/yr of ammonia to Europe by 2060 on the other corridor

WHAT THE IMPORTERS BUY IT FOR

Korea	Hydrogen and ammonia for power generation and steelmaking
Japan	The same, under a 10–15 year support programme
Singapore	Ammonia specifically for maritime bunkering — the leading fuel hub

Note DNV's caution: Hydrogen is not the next LNG. It will be produced in every region, so trade complements local production rather than replacing it — long-distance pure hydrogen trade stays constrained by conversion losses and volumetric density. The volume is in ammonia.

The Fuel Competition

Three contenders, three different risk profiles — and the design decision comes before the answer is known

LNG

The incumbent alternative

- ~1,000 vessels on order today
- Mature bunkering and supply chain
- Methane slip exposed under FuelEU well-to-wake accounting
- A transition fuel, not an endpoint

e-Methanol

The early mover

- ~10-year track record as a marine fuel
- Liquid at ambient — simplest retrofit
- Requires biogenic or captured CO_2
- That CO_2 is the ceiling on scale

Ammonia

The long-term winner in DNV's view

- No carbon source required
- Existing global trade and storage
- Toxicity, engine readiness, safety standards
- Barriers expected resolved early-to-mid 2030s

And they are not the only pathways — DNV lists LPG, growing biofuels, onboard carbon capture and nuclear propulsion alongside them.

Key Messages

1. **Shipping cannot decarbonise without hydrogen**

   Direct electrification is structurally limited — batteries suit ferries, not deep sea.

2. **Derivatives, not pure hydrogen, carry the transition**

   Maritime demand arises almost exclusively through ammonia and methanol; hydrogen is cargo, not bunker.

3. **Regulation sets the pace, not fuel price**

   Clean hydrogen is not market competitive; the cost gap closes through carbon pricing and mandates.

4. **Ammonia overtakes methanol in the early 2040s**

   Barriers — toxicity, engine readiness, safety standards — are expected to be resolved early-to-mid 2030s.

5. **Fuel-capable, not fuel-ready, is what counts**

   Only fuel-capable ships generate demand. The design decision precedes the final rule text.

Hydrogen Roadmap for Shipping

2025 – 2030

- Efficiency measures and biofuels
- Regulatory framework confirmed
- e-Methanol newbuilds and retrofits

2030 – 2040

- Ammonia barriers resolved
- Bunkering hubs consolidate
- Asia-Pacific corridors build out

2040 – 2060

- Ammonia the dominant derivative
- ~15 % of world hydrogen demand
- Deep decarbonisation of shipping

“Hydrogen is not the fuel of everything — but it will be one of the most important fuels for shipping. The future of maritime decarbonisation will be built on hydrogen derivatives, above all ammonia and e-methanol.”

DNV

WHEN TRUST MATTERS

ENERGY TRANSITION OUTLOOK 2026 HYDROGEN TO 2060

From pilots to scale for renewable and low-carbon hydrogen and synthetic fuels

Thank you for your attention.

chang-woo.son@dnv.com
+821040088770

www.dnv.com